

DISASTER DETECTION

Submitted by,

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Under the guidance of,

Dr. SAMPATH A K

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION SCIENCE AND TECHNOLOGY

(Artificial Intelligence & Data Science)

At



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PRESIDENCY UNIVERSITY

BENGALURU

MAY 2025

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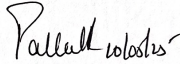
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CERTIFICATE

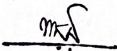
This is to certify that the Internship/Project report “**DISASTER DETECTION**” being submitted by “A SHABARRIESH, A V ROHITH KUMAR, M B V SANDEEP REDDY” bearing roll number “20211ISD0030, 20211ISD0009, 20211ISD0035” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in Information Science and Technology is a Bonafide work carried out under my supervision.



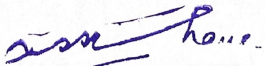
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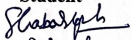
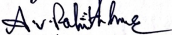
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DECLARATION

I hereby declare that the work, which is being presented in the report entitled “**DISASTER DETECTION**”, in partial fulfillment for the award of the Degree of **Bachelor of Technology in Information Science and Technology**, is a record of my own investigations carried out under the guidance of **Dr. SAMPATH A K, Professor, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.**

I further declare that the matter presented in this report has not been submitted by me anywhere else for the award of any other Degree.

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ABSTRACT

Natural disasters such as hurricanes and earthquakes cause widespread destruction, significantly impacting lives and economies. Accurate and timely prediction of these disasters is critical for effective preparedness and mitigation efforts. This project develops a **dual-disaster prediction system** leveraging **machine learning models** to forecast hurricanes and earthquakes using historical and real-time data. A **Random Forest model** is implemented for hurricane prediction due to its robustness in handling high-dimensional meteorological data, while **Logistic Regression** is employed for earthquake prediction, excelling in binary classification tasks. The system integrates data preprocessing, predictive modeling, and user-friendly visualization tools to provide actionable insights. Real-time monitoring capabilities enable the system to deliver early warnings via automated alerts, empowering authorities and communities to take proactive measures. The solution also includes a responsive web-based interface designed for intuitive interaction and accessible visualizations. By enhancing the accuracy and efficiency of disaster prediction, this framework aims to reduce disaster-related losses, safeguard lives, and support informed decision-making for emergency management teams and policymakers.

ACKNOWLEDGEMENT

First of all, we indebted to the **GOD ALMIGHTY** for giving me an opportunity to excel in our efforts to complete this project on time.

We express our sincere thanks to our respected dean **Dr. Md. Sameeruddin Khan**, Pro-VC - Engineering and Dean, Presidency School of Computer Science and Engineering & Presidency School of Information Science, Presidency University for getting us permission to undergo the project.

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We would like to convey our gratitude and heartfelt thanks to the PIP4004 University Project Coordinator **Mr. Md Ziaur Rahman** and **Dr. Sampath A K**, department Project Coordinators "**Mr Srinivas mishra**" and Git hub coordinator **Mr. Muthuraj**.

We thank our family and friends for the strong support and inspiration they have provided us in bringing out this project.

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